

Chenyi Zhu

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EDUCATION

University of California, San Diego (UCSD)

La Jolla, CA

B.S. Double Major: Data Science & Applied Mathematics **GPA: 3.96/4.00**

Sep. 2024 – Present

- Courses: Data Structures & Algorithms, Linear Algebra, Probability, Machine Learning, Optimization

EXPERIENCE

Autonomous Systems Research Intern

Jun. 2025 – Sep. 2025

State Key Lab of AI Unmanned Systems, Tongji University

Shanghai, China

- Set up and calibrated an optical motion-capture system; placed/verified markers and ran capture sessions to collect humanoid motion data.
- Processed mocap trajectories with gap filling and filtering; labeled joints and time-synchronized motion sequences with joint logs for consistent timestamps and sampling rates.
- Retargeted mocap motion to simulation skeletons for imitation-learning datasets; built joint-mapping rules (DoF alignment, joint limits, coordinate frames) and validated outputs via replay.
- Integrated retargeted motions into **NVIDIA Isaac Sim**: imported robot/skeleton assets (USD/URDF), configured articulations and physics timestep, replayed reference motions, and tuned controller gains/hyper-parameters; maintained Python scripts/configs and run logs for reproducibility.

Matlab Tutor (MATH 18: Linear Algebra)

Mar. 2025 – Jun. 2025

UC San Diego

La Jolla, CA

- Supported students on MATLAB-based components of MATH 18 (homework and practice problems); debugged common issues and clarified assignment requirements.
- Guided implementation of core linear-algebra computations in MATLAB: solving linear systems (row reduction), matrix operations, determinants/inverses, and eigenvalues/eigenvectors.
- Helped students connect theory to computation (subspaces, orthogonality/least squares) by verifying results numerically and interpreting outputs.

PROJECTS

Footstep-Guided Humanoid Obstacle Avoidance | PPO (Actor-Critic), PyTorch

Jun. 2025 – Present

- Implemented an actor-critic policy in PyTorch (actor/critic modules + shared base API) for continuous humanoid control with footstep-conditioned observations.
- Built PPO training pipeline: rollout collection, advantage/value targets, policy/value losses with entropy bonus, and logging for stability (losses, KL/entropy trends).
- Integrated in Isaac Sim for closed-loop rollouts and benchmarking; evaluated success/collision/fall metrics and tuned reward/termination to improve robustness.

Imaging Research in Denoising and Segmentation

Sep. 2024 – Sep. 2025

Mentored Research (Tongji University)

Remote / Shanghai, China

- Built variational models for image denoising and segmentation using quadratic and total-variation (TV) regularization.
- Derived Euler-Lagrange optimality conditions and implemented gradient-based and primal-dual solvers.
- Produced a technical report with reproducible code and experiments; time commitment: 4 hrs/week (36 weeks/year).

Mathematical Modeling Research on Airport Taxi Decisions

Independent Research

Shanghai, China

- Studied taxi-driver dispatch decisions at Shanghai Hongqiao Airport (e.g., join/skip a queue, switch waiting areas) and organized options into an analytic hierarchy to compare strategies consistently.
- Built a multi-server FCFS queueing model for taxi boarding with Poisson arrivals and exponential service; analyzed how arrival/service rates and server counts affect waiting time and queue length.
- Defined a unit-time revenue model combining fare assumptions, expected trip duration, and idle/wait time to estimate earnings rate across strategies under uncertainty.
- Implemented MATLAB simulations to validate analytic assumptions and ran sensitivity analyses on key parameters (arrival rate, service rate, fare/trip-time); wrote a formal research report with results and recommendations.

TECHNICAL SKILLS

Languages: Java, Python, SQL, R, Matlab

Spoken Languages: Chinese (Native), English (Fluent)

Developer Tools: Git, Docker, TravisCI, Google Cloud Platform, VS Code, PyCharm,

Libraries: NumPy, pandas, Matplotlib, SciPy, scikit-learn, PyTorch, TensorFlow, OpenCV, SymPy, CVXPY, Stable-Baselines3